

Paper 20 - Working English translation

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THE WEIL CONJECTURE I

PIERRE DELIGNE

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[Polished translation of the opening paragraphs]

In this paper Deligne proves the Weil conjecture on the eigenvalues of Frobenius endomorphisms. A precise statement is given in (1.6). The argument is presented in as geometric and elementary a form as possible, with many reminders of background material. The genuinely new ingredients are the results of sections 3, 6, 7, and 8.

A sequel develops refinements of the intermediate results and applications, among them the hard Lefschetz theorem for repeated cup-product with the cohomology class of a hyperplane section.

The text follows closely six lectures delivered in Cambridge in July 1973. Deligne thanks Nicholas Katz for permission to use his notes.

[Source-aligned literal body continues at Section 1]

1. Grothendieck's theory: cohomological interpretation of L-functions

(1.1) Let X a schema of type fini on Z , $|X|$ l'ensemble of points fermes of X and, for $x \in |X|$, let $N(x)$ the nombre d'elements of the corps residuel $k(x)$ of X en x . La zeta function of Hasse-Weil of X is

$$(1.1) \quad z_X(s) = \prod_{x \in |X|} (1 - N(x)s)^{-1}$$

(this product converge absolument for $R(s)$ assez grand). For $X = \text{Spec}(Z)$, $z_X(s)$ is the zeta function of Riemann.

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Nous considererons exclusivement the case or X is a schema on a finite field F_q .

For $x \in |X|$, we ecrivons q_x plutot que $N(x)$. Posant $\deg(x) = [k(x) : F_q]$, $q_x = q^{\deg(x)}$. Il y a interet a introduire the variable $t = q^{-s}$. Posons

$$(1.2) \quad Z(X_0; t) = \prod_{x \in |X_0|} (1 - t^{\deg(x)})^{-1};$$

this product infini converge for $|t|$ assez petit, and one a

$$(1.3) \quad z_X(s) = Z(X_0; q^{-s}).$$

(1.2) Dwork (On the rationality of the zeta function of an algebraic variety, Amer. J. Math., 82, 1960, p. 631-648) and Grothendieck ([1] and SGA 5) ont demontre que $Z(X, t)$ is a fonction rationnelle of t .

For Grothendieck, c'est the a corollary of results generaux en cohomologie ℓ -adique (ℓ designe a nombre premier different of the caracteristique p of F_q). Ceux-ci fournissent a interpretation cohomologique of zeros and of poles of $Z(X; t)$, and a functional equation lorsque X is proper and smooth. Les methodes of Dwork are p -adiques. For X a nonsingular hypersurface in l'espace projectif, elles lui fournissaient egalement a interpretation cohomological of zeros and of poles, and l'functional equation. Elles ont inspire the theorie cristalline of Grothendieck and Berthelot, qui for X proper and smooth fournit a interpretation cohomologique p -adique of zeros and poles, and l'equation fonctionnelle. Se basant on of idees of Washnitzer, Lubkin a cree a variante of cette theorie, valable seulement for X proper, smooth and releuable en caracteristique zero (A p -adic proof of Weil's conjectures, Ann. of Math., 87, 1968, p. 105-255).

Nous ferons a usage essentiel of results of Grothendieck, and the let us recall ci-dessous.

(1.3) Let X a algebraic variety on a corps algebraically closed k of caracteristique p , i.e. a schema separe of type fini on k . On n'exclut pas the case $p = 0$. For all nombre premier $\ell \neq p$, Grothendieck a defini of cohomology groups ℓ -adiques $H^i(X, Q_\ell)$. Il a aussi defini of groups of cohomology with proper support $H^i_c(X, Q_\ell)$. For X proper, the deux coincident. Les $H^i(X, Q_\ell)$ are of spaces vectoriels of dimension finie on Q_ℓ , zero for $i > 2 \dim(X)$.

(1.4) Let X_0 a algebraic variety on F_q , F_q a cloture algebrique of F_q and X the algebraic variety on F_q deduite of X_0 par extension of scalaires of F_q a F_q . Dans the langage of Weil or Shimura, one

exprimerait cette situation par : "Let X a algebraic variety definie on F_q ". Let $F : X \rightarrow X$ the Frobenius morphism ; il envoie the point of coordonnees x vers the point of coordonnees xq ; en d'autres termes, for $U \subset X$ a ouvert of Zariski of X_0 , definissant a ouvert U of X , one a $F^{-1}(U) = U$; for $x \in H^0(U, \mathcal{O})$, one a $F * x = xq$. Identifions l'ensemble

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$|X|$ of closed points of X a $X_0(F_q)$ (l'ensemble $\text{Hom}_{F_q}(\text{Spec}(F_q), X_0)$ of points of X_0 a coefficients in F_q), and let $ph \in \text{Gal}(F_q/F_q)$ the Frobenius substitution : $ph(x) = xq$. L'action of F on $|X|$ s'identifie a l'action of ph on $X_0(F_q)$. Des lors :

a) L'ensemble X^F of closed points of X fixes sous F s'identifie a l'ensemble $X_0(F_q) \subset X_0(F_q)$ of points of X definis on F_q . Ceci exprime simplement que, for $x \in F_q$, one a $xq = x$ if and seulement if $x \in F_q$.

b) De meme, l'ensemble X^F of closed points of X fixes sous the nieme itere of F s'identifie a $X_0(F_{q^n})$.

c) L'ensemble $|X_0|$ of closed points of X_0 s'identifie a l'ensemble $|X|/F$ of orbites of F (or of ph) in $|X|$. Le degre $\deg(x)$ of $x \in |X_0|$ is the nombre d'elements of l'orbite correspondante.

d) De b) and c) resulte the formule

$$(1.4) \quad \#X^F = \#X_0(F_{q^n}) = \sum_{x \in |X_0|} \frac{\deg(x)}{\deg(x)|n}$$

(for $x \in |X_0|$ and $\deg(x) \mid n$, x defines $\deg(x)$ points a coordonnees in F_{q^n} , all conjugues on F_q).

(1.5) Le morphism F is fini, en particulier proper. Il induit therefore of maps $F^* : H_{\text{ci}}(X, \mathbb{Q}) \rightarrow H_{\text{ci}}(X, \mathbb{Q})$. Grothendieck a proves the formule of Lefschetz

$$\#X^F = \sum_i (-1)^i \text{Tr}(F^*, H_{\text{ci}}(X, \mathbb{Q}))$$

the membre of droite, a priori a nombre $\mathbb{Q}$ -adique, is entier, and egal au membre of gauche. On notera qu'une telle formule n'est raisonnable que parce que $dF = 0$, meme a l'infini (X n'est pas suppose proper) ; the relation $dF = 0$ implique que the points fixes of F are of multiplicity a.

Une formule analogue vaut for the iterates of F :

$$(1.5) \quad \#X^F = \#X_0(F_{q^n}) = \sum_i (-1)^i \text{Tr}(F^{*n}, H_{\text{ci}}(X, \mathbb{Q}))$$

Prenons the derivee logarithmique of (??) :

$$(1.6) \quad \frac{d}{dt} \log Z(X_0, t) = \sum_{x \in |X_0|} \frac{-t \deg(x)}{1 - t \deg(x)} = \sum_{x \in |X_0|} \sum_{n \geq 0} \frac{\deg(x)^n}{\deg(x) t^n} = \sum_{n \geq 0} \#X_0(F_{q^n}) t^n$$

For F a endomorphism d'un space vectoriel V , one a a identite of series formelles

$$(1.7) \quad \frac{d}{dt} \log(\det(1 - F t, V) - 1) = \sum_{n \geq 0} \text{Tr}(F^n, V) t^n$$

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(the verifier for $\dim(V) = 1$, and observer que the deux membres are additifs en V in a sequence exacte courte). Substituant (??) in (??) and appliquant (??), one trouve

$$\frac{d}{dt} \log Z(X_0, t) = \sum_i (-1)^i \frac{d}{dt} \log \det(1 - F * t, H_{\text{ci}}(X, \mathbb{Q})) - 1$$

let

$$Y = \sum_i \det(1 - F * t, H_{\text{ci}}(X, \mathbb{Q})) (-1)^{i+1}$$

$$(1.8) \quad Z(X_0, t) = \dots$$

Le membre of droite is a element of $\mathbb{Q}[[t]]$. La formule affirme que son developpement of Taylor en $t = 0$, a priori a serie formelle in $\mathbb{Q}[[t]]$ of terme constant a, is in $\mathbb{Z}[[t]]$, and is egal au membre of gauche, lui aussi considers as a serie formelle en t. Cette formule is l'interpretation cohomological of Grothendieck of the fonction Z.

Notre result principal is the suivant.

Theoreme 1.1 (1.6). Let X_0 a variety projective nonsingular (=smooth) on $\mathbb{F}_q$. For chaque i, the polynome caracteristique $\det(1 - F^* t, H^i(X, \mathbb{Q}))$ is a coefficients entiers independants of q ($q \neq p$). Les roots complexes a of ce polynome (the conjugues complexes of eigenvalues of F^*) are of value absolue $|a| = q^{i/2}$.

Montrons already que ?? resulte of the result apparemment more faible suivant.

Lemme 1.2 (1.7). For chaque i, and chaque $q \neq p$, the eigenvalues of l'endomorphism F^* of $H^i(X, \mathbb{Q})$ are of nombres algebriques dont all the conjugues complexes a satisfy $|a| = q^{i/2}$.

Preuve of (??) $\Rightarrow$ (??). -- Regardons $Z(X_0, t)$ as a P serie for-

melle of terme constant 1, element of $\mathbb{Z}[[t]]$: $Z(X_0, t) = \sum_{n=0}^{\infty} a_n t^n$.

After (??), l'image of $Z(X_0, t)$ in $\mathbb{Q}[[t]]$ is the developpement of Taylor d'une fraction rationnelle. Ceci signifie que for N and M assez grands ($\geq$ the degres of numerateurs and denominateurs) the determinants of Hankel

$$H_k = \det((a_{i+j+k})_{0 \leq i, j \leq M}) \quad (k > N)$$

are zero. Cette nullite is vraie in $\mathbb{Q}$ if and seulement if elle l'est in $\mathbb{Q}$; $Z(X_0, t)$ is therefore the developpement in serie of Taylor d'un element of $\mathbb{Q}(t)$. En d'autres termes,

$$Z(X_0, t) \in \mathbb{Q}(t) \cap \mathbb{Z}[[t]].$$

Ecrivons $Z(X_0, t) = P/Q$, with $P, Q \in \mathbb{Z}[t]$, premiers entre eux, and of terme constant positif. After a lemma of Fatou, que $Z(X_0, t)$ let in $\mathbb{Z}[[t]]$ and of terme constant a implique que the termes constants of P and Q are 1. Posons

$$P_i(t) = \det(1 - F^* t, H^i(X, \mathbb{Q})).$$

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After l'hypothese (??), the P_i are premiers entre eux. Le membre of droite of (??) is therefore in the form irreductible, and

$$P(t) = \prod_{i \text{ impair}} P_i(t), \quad Q(t) = \prod_{i \text{ pair}} P_i(t).$$

Let K the sous-corps d'une cloture algebrique $\mathbb{Q}$ of $\mathbb{Q}$ engendre on $\mathbb{Q}$ par the roots of $R(t) = P(t)Q(t)$. Les roots of $P_i(t)$ are celles of roots of $R(t)$ ayant the property que all leurs conjugues complexes are of value absolue $q^{-i/2}$. Cet ensemble is stable sous $\text{Gal}(K/\mathbb{Q})$. Le polynome $P_i(t)$ is therefore a coefficients rationnels. After the lemma of Gauss (or parce que the roots of P_i , being roots of $R(t)$, are of inverses d'entiers algebriques), il is meme a coefficients entiers. La description ci-dessus of roots of $P_i(t)$ is independante of q ; the polynome $P_i(t)$ lui-meme is therefore independant of q .

La sequence of cet article is consacree a the proof of (??).

(1.8) The theory of Grothendieck fournit a interpretation cohomologique non seulement of fonctions zeta, mais again of L-functions. Les results are the suivants.

(1.9) Let X a algebraic variety on a corps k. For the definition d'un $\mathbb{Q}$ -sheaf constructible on X, je renvoie a SGA 5 VI. Qu'il suffice of dire que :

a) If F is a $\mathbb{Q}$ -sheaf constructible on X, il existe a partition finie of X en parties localement fermees X_i telles que $F|_{X_i}$ let constant tordu.

b) Supposons X connexe and let x a point geometric of X. For F constant tordu, $\pi_1(X, x)$ agit on the fibre F_x ; the functor fibre en x is a equivalence of categorie

($\mathbb{Q}$ -sheaves constructibles constants tordus on X) - (representations continues of $\pi_1(X, x)$

Une telle representation ne se factorise pas, en general, a travers a quotient fini of $\pi_1(X, x)$.

c) If $k = \mathbb{C}$, the $\mathbb{Q}$ -sheaves constructibles on X s'identifient aux sheaves of $\mathbb{Q}$ -spaces vectoriels F on X an, tels qu'il existe a partition finie of X en parties Zariski-localement fermees X_i , and for chaque i a systeme local of $\mathbb{Z}$ -Modules libres of type fini L_i on X_i , with $F|_{X_i} = L_i \otimes_{\mathbb{Z}} \mathbb{Q}$.

Nous ne considererons que of $\mathbb{Q}$ -sheaves constructibles, and the appellerons simplement $\mathbb{Q}$ -sheaves.

(1.10) Supposons k algebraically closed, and let F a $\mathbb{Q}$ -sheaf on X . Grothendieck a defini of cohomology groups $H^i(X, F)$ and $H_{ci}(X, F)$. Les $H_{ci}(X, F)$ are of spaces vectoriels of dimension finie on $\mathbb{Q}$, zero for $i > 2 \dim(X)$.

For $k = \mathbb{C}$, the $H^i(X, F)$ and $H_{ci}(X, F)$ are the groups of cohomologie usuels (resp. support proper) of X an, a coefficients in F .

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(1.11) Let X_0 a algebraic variety on F_q , X the variety on F_q correspondante, and F_0 a sheaf d'ensembles on X_0 (for the topologie etale). On note F son image reciproque on X . Outre the morphism of Frobenius $F : X \rightarrow X$, one dispose then d'un isomorphism canonical

$\tilde{F} : F * F \rightarrow F$. En voici a description. On regarde F_0 as a space etale on X_0 , i.e. one identifie F_0 a l'espace algebrique $[F_0]$, muni d'un morphism etale $f : [F_0] \rightarrow X_0$, tel que F_0 let the sheaf of sections locales of $[F_0]$. L'espace analogue $[F]$, etale on X , se deduit of $[F_0]$ par extension of scalaires. On dispose therefore d'un diagramme commutatif

$$\begin{array}{ccc} [F] & & / [F] \\ f \downarrow & & \downarrow f \\ X & \square & \square \\ & F & /X \end{array}$$

d'ou a morphism $[F] \rightarrow X \times F$ $[F] = [F * F]$, qui is a isomorphism

car f is etale. Son inverse defines l'isomorphisme $F * F \rightarrow F$ cherche.

Cette construction se generalise aux $\mathbb{Q}$ -sheaves.

(1.12) Let X_0 a algebraic variety on F_q , F_0 a $\mathbb{Q}$ -sheaf on X_0 , (X, F) deduit of (X_0, F_0) par extension of scalaires of F_q a F_q ,

$F : X \rightarrow X$ and $\tilde{F} : F * F \rightarrow F$. Le morphism fini F and $\tilde{F}$ definissent a endomorphism

$$F^* : H_{ci}(X, F) \rightarrow H_{ci}(X, F).$$

For $x \in |X|$, F^* defines a morphism $Fx^* : F_F(x) \rightarrow F_x$. For $x \in X \times F$, c'est a endomorphism of F_x . Grothendieck a demontre the formule of Lefschetz

$$\sum_{x \in X \times F} \text{Tr}(F^* x^*, F_x) = \sum_i (-1)^i \text{Tr}(F^*, H_{ci}(X, F)).$$

Une formule analogue vaut for the iterates of F : the nieme itere of F^* defines of morphisms $F^{*n} : F_F^n(x) \rightarrow F_x$; for x fixe sous F^n , F^{*n} is a endomorphism, and

$$(1.9) \quad \sum_{x \in X \times F^n} \text{Tr}(F^{*n} x^*, F_x) = \sum_i (-1)^i \text{Tr}(F^{*n}, H_{ci}(X, F)).$$

(1.13) Let $x_0 \in |X_0|$, Z l'orbite correspondante of F in $|X|$ and $x \in Z$. L'orbite Z a $\deg(x_0)$ elements (1.4). On note Fx_0 l'endomorphism $* \deg(x)$

F_x of F_x , and one pose $\det(1 - Fx_0 t, F_0) = \det(1 - Fx^* \deg(x) t, F_x)$.

A isomorphism pres, (F_x, Fx_0) ne depend pas of the choix of x . Ceci justifie d'omettre x of the notation. On utilisera a notation analogue for d'autres fonctions of (F_x, Fx_0) .

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(1.14) Definissons $Z(X_0, F_0, t) \in \mathbb{Q}[[t]]$ par the product

$$(1.10) \quad Z(X_0, F_0, t) = \prod_{x \in |X_0|} \det(1 - Fx t \deg(x), F_0)^{-1}.$$

For F the sheaf constant $\mathbb{Q}$, one retrouve (??). After (??), the derivee logarithmique of Z is

$$(1.11) \quad \frac{d}{dt} \log Z(X_0, F_0, t) = \sum_n \frac{1}{n} \sum_{x \in X} \text{Tr}(F^n x^*, F_x) t^{n-1} = \sum_{x \in X} \text{Tr}(F^n x^*, F_x) t^{n-1}.$$

Substituant (??) in (??), one trouve par the meme calcul qu'en (1.5) the generalisation suivante of (??)

$$(1.12) \quad Z(X_0, F_0, t) = \prod_{i=1}^Y \det(1 - F^* t, H_{ci}(X, F))^{(-1)^{i+1}}.$$

Cette formule is a identite in $\mathbb{Q}[[t]]$.

(1.15) Il is sometimes commode d'utiliser a langage galoisien plutot que geometric. Voici the dictionnaire.

If F_1 and F_2 are deux clotures algebriques of F_0 , (X_1, F_1) on F_0 defines par extension of scalaires (X_1, F_1) on F_0 and (X_2, F_2) on F_0 .
 $F_1 \sim F_2$

Tout F_0 -isomorphism $s : F_1 \rightarrow F_2$ induit a isomorphism

$$H^i(X_1, F_1) \rightarrow H^i(X_2, F_2).$$

En particulier, for $F_0 = F_0$ (note F_0), one trouve que $\text{Gal}(F_0/F_0)$ agit on $H^i(X, F)$ (action par transport of structure). Let ph the substitution of Frobenius. On satisfies que

$$F^* = \text{ph}^{-1}$$

(in $\text{End}(H^i(X, F))$).

Ceci amene a define the Frobenius geometric $F \in \text{Gal}(F_0/F_0)$ as being ph^{-1} . On a

$$(1.13) \quad F^* = F.$$

Let x a point geometric of X_0 , localise en $x_0 \in |X_0|$. Par transport of structure, the group $\text{Gal}(k(x)/k(x_0))$ agit on the fibre $(F_0)_x$ of F_0 en x ; en particulier, the Frobenius geometric relatif $F_x \in \text{Gal}(k(x)/k(x_0))$, agit. For x defini par a point ferme, again note x , of X , one a $(F_0)_x = F_x$, and

$$(1.14) \quad F_x = F_x \cdot \deg(x) \quad (\text{in } \text{End}(F_x)).$$

En notations galoisiennes, (??) s'ecrit

$$\det(1 - F_x t \deg(x), F_0) = \prod_{x \in |X_0|} \det(1 - F t, H^i(X, F))^{i+1} (-1)^i.$$

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2. Grothendieck's theory: Poincare duality

(2.1) For expliquer the relation entre roots of l'unite and orientations, je vais au prealable redire in a langage farfelu deux case classiques.

a) Varieties differentiables. -- Let X a variety differentiable purement of dimension n . Le sheaf d'orientation Z_0 on X is the sheaf localement isomorphe au sheaf constant Z , dont the sections inversibles on a ouvert U of X correspond aux orientations of U . Une orientation of X is a isomorphism of Z_0 on the sheaf constant Z . La classe fondamentale of X is a morphism $\text{Tr} : H^{2n}(X, Z_0) \rightarrow Z$; if X is orientee, il s'identifie a a morphism $\text{Tr} : H^{2n}(X, Z) \rightarrow Z$. La Poincare duality s'exprime a l'aide of the classe fondamentale.

b) Varieties complexes. -- Let C a cloture algebrique of $\mathbb{R}$. Une variete algebrique complexe smooth, or plutot the variety differentiable sous-jacente, is always orientable. For l'orienter, il suffices d'orienter C lui-meme. Ceci revient au choix :

- a) a choisir l'une of deux roots of l'equation $X^2 = -1$; one l'appelle $+i$;
- b) a choisir a isomorphism of $\mathbb{R}/Z$ with $U(1) = \{z \in \mathbb{C} \mid |z| = 1\}$; $+i$ is l'image of $1/4$;
- c) a choisir l'un of deux isomorphisms $x \mapsto \exp(+2\pi i x)$ of $\mathbb{Q}/Z$ on the group of roots of l'unite of C , qui se prolonge par continuite en a isomorphism of $\mathbb{R}/Z$ on $U(1)$.

On note $Z(1)$ a Z -module libre of rang a dont l'ensemble a deux elements of generateurs let en correspondance canonical with l'un of ensembles a deux elements a), b), c). Le more simple is of prendre $Z(1) = \text{Ker}(\exp : C \rightarrow C^*)$. Au generateur $y = +2\pi i$ corresponds l'isomorphisme c) : $x \mapsto \exp(xy)$.

Let $Z(r)$ the puissance tensorielle r ieme of $Z(1)$. If X is a variete algebrique complexe smooth purement of dimension complexe r , the sheaf d'orientation of X is the sheaf constant of value $Z(r)$.

(2.2) For "orienter" the algebraic varieties on k algebraically closed of caracteristique 0 , il faut choisir a isomorphism of $\mathbb{Q}/Z$ on the group of roots of l'unite of k . L'ensemble of ces isomorphisms is a space principal homogene sous Z^* (non more sous Z^*). Lorsqu'on s'interesse seulement a the cohomology ℓ -adique, il suffices of considerer the roots of l'unite d'ordre a puissance of ℓ , and of supposer the caracteristique p of k differente of ℓ . On note $Z/\ell^n(1)$ the group of roots of l'unite of k d'ordre divisant ℓ^n . For n variable, the $Z/\ell^n(1)$

formant a systeme projectif, d'applications of transition the

$$Z^{n+1, n} : Z/\ell^{n+1}(1) \rightarrow Z/\ell^n(1) : x \mapsto x\ell = x\ell^n.$$

On pose $Z_\ell(1) = \lim Z/\ell^n(1)$ and $Q_\ell(1) = Z_\ell(1) \otimes_{Z_\ell} Q_\ell$.

On note $Q_\ell(r)$ the puissance tensorielle r ieme of $Q_\ell(1)$; for $r \in \mathbb{Z}$, r negatif, one pose again $Q_\ell(r) = Q_\ell(-r)$.

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En tant qu'espace vectoriel on Q_ℓ , $Q_\ell(1)$ is isomorphe a Q_ℓ . Toutefois the group of automorphismes of k agit non trivialement on $Q_\ell(1)$: il agit via the caractere $\text{kh} : \text{Gal}(k/k) \rightarrow \mathbb{Z}^*$ a values in $\mathbb{Z}^*$ qui gives son action on the roots of l'unite. En particulier, if $k = \mathbb{F}_q$, the Frobenius substitution $\text{ph} : x \mapsto x^q$ agit par multiplication par q .

Let X a algebraic variety smooth purement of dimension n on k . Le sheaf d'orientation of X en cohomologie ℓ -adique is the Q_ℓ -sheaf constant $Q_\ell(n)$. La classe fondamentale is a morphism

$$\text{Tr} : \text{Hc}2n(X, Q_\ell(n)) \rightarrow Q_\ell,$$

let again

$$\text{Tr} : \text{Hc}2n(X, Q_\ell) \rightarrow Q_\ell(-n).$$

Theoreme 2.1 (2.3). (Poincare duality). For X proper and smooth purement of dimension n , the form bilineaire

$$\text{Tr}(x \cup y) : \text{H}^i(X, Q_\ell) \otimes \text{H}^{2n-i}(X, Q_\ell(n)) \rightarrow Q_\ell(-n)$$

is a dualite parfaite (elle identifie $\text{H}^i(X, Q_\ell)$ au dual of $\text{H}^{2n-i}(X, Q_\ell(n))$).

(2.4) Let X_0 a algebraic variety proper and smooth on $\mathbb{F}_q$, purement of dimension n , and X on $\mathbb{F}_q$ deduit of X_0 par extension of scalaires. Le morphism (??) is compatible a l'action of $\text{Gal}(\mathbb{F}_q/\mathbb{F}_q)$. If the a_j are the eigenvalues of the Frobenius geometric F agissant on $\text{H}^i(X, Q_\ell)$, the eigenvalues of F agissant on $\text{H}^{2n-i}(X, Q_\ell)$ are therefore the $(q^n a_{j-1})$.

(2.5) Supposons for simplifier X connexe. La proof of (??) se transpose as suit in a langage geometric, plutot que galoisien (cf. (1.15)).

- Le cup-produit met $\text{H}^i(X, Q_\ell)$ and $\text{H}^{2n-i}(X, Q_\ell)$ en dualite parfaite a values in $\text{H}^{2n}(X, Q_\ell)$, qui is of dimension a .
- Le cup-produit commute a l'image reciproque F^* par the morphisme of Frobenius $F : X \rightarrow X$.
- Le morphism F is fini and of degre q^n : on $\text{H}^{2n}(X, Q_\ell)$, F^* is the multiplication par q^n .
- Les eigenvalues of F^* ont therefore the property (2.4).

(2.6) On pose $\text{kh}(X) = \sum_i (-1)^i \dim \text{H}^i(X, Q_\ell)$. If n is impair, the form $\text{Tr}(x \cup y)$ on $\text{H}^n(X, Q_\ell)$ is alternee; l'entier $n\text{kh}(X)$ is therefore always pair. On deduit facilement of (??) and of (??), (2.4) que

$$Z(X_0, t) = e \cdot q^{n\text{kh}(X)/2} \cdot \text{tkh}(X) \cdot Z(X_0, q^{-n}t^{-1})$$

or $e = \pm 1$. If n is impair, notons N the multiplicite of the value proper $q^{n/2}$ of F^* agissant on $\text{H}^n(X, Q_\ell)$ (i.e. the dimension of the sous-espace proper generalise correspondant). On a

$$e = \begin{cases} 1 & \text{if } n \text{ is impair,} \\ (-1)^N & \text{if } n \text{ is pair.} \end{cases}$$

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C'est the the formulation of Grothendieck of l'functional equation of fonctions Z .

(2.7) Nous aurons besoin d'autres forms of the theorem of dualite. Le case of courbes we suffirait in celles-ci. If F is a Q_ℓ -sheaf on a algebraic variety X on k algebraically closed, we noterons $F(r)$ the sheaf $F \otimes_{Q_\ell} Q_\ell(r)$. Ce sheaf is (non canonically) isomorphe a F .

Theoreme 2.2 (2.8). Supposons X smooth purement of dimension n and F constant tordu. Let F the dual of F . La form bilineaire

$$\text{Tr}(x \cup y) : \text{H}^i(X, F) \otimes \text{Hc}2n-i(X, F(n)) \rightarrow \text{Hc}2n(X, Q_\ell(n)) \rightarrow Q_\ell$$

Tr

is a dualite parfaite.

(2.9) Supposons X connexe and let x a point ferme of X . Le functor $F \mapsto F_x$ is a equivalence of the categorie of Q_ℓ -sheaves constants tordus with celle of representations ℓ -adiques of $\pi_1(X, x)$. Via cette equivalence, $\text{H}^0(X, F)$ s'identifie aux invariants of $\pi_1(X, x)$ agissant in F_x :

$$(2.1) \quad \text{H}^0(X, F) = F_x^{\pi_1(X, x)}.$$

After (??), for X smooth and connexe of dimension n , one a therefore

$$\text{Hc}2n(X, F) = \text{H}^0(X, F(n)) = ((F(n))^{\pi_1(X, x)}).$$

La dualite echange invariants (more grand sous-espace invariant) and coinvariants (more grand quotient invariant). Cette formule se recrit therefore

$$Hc2n(X, F) = (F_x)_{n1}(X, x) (-n).$$

Nous ne l'utiliserons que for $n = 1$.

Scholie 2.3 (2.10). Let X a courbe smooth and connexe on k algebriquement clos, x a point ferme of X and F a $\mathbb{Q}$ -sheaf constant tordu. On a :

(i) $Hc0(X, F) = 0$ if X is affine.

(ii) $Hc2(X, F) = (F_x)_{n1}(X, x) (-1)$.

L'assertion (i) signifie simplement que F n'a pas of section a support fini.

(2.11) Let X a courbe projective, smooth and connexe on k algebriquement clos, U a ouvert of X , complement d'un ensemble fini S of closed points of X , j l'inclusion $U \rightarrow X$ and F a $\mathbb{Q}$ -sheaf constant tordu on U . Let $j_! F$ the $\mathbb{Q}$ -sheaf constructible image directe of F . Sa fibre en $x \in S$ is of rang inferieur or egal au rang of sa fibre en a point general ; c'est a space d'invariants sous a group of monodromie locale.

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Theoreme 2.4 (2.12). La form bilineaire

$$\text{Tr}(xuy) : H^i(X, j_! F) \otimes H^{2-i}(X, j_* F(1)) \rightarrow H^2(X, j_! F \otimes j_* F(1)) \rightarrow H^2(X, \mathbb{Q}(1)) \xrightarrow{\text{Tr}}$$

is a dualite parfaite.

(2.13) Il we will be commode of disposer of $\mathbb{Q}$ -sheaves $\mathbb{Q}(r)$ on a quelconque schema X or ` is inversible. Le all is of define the $Z/n(1)$. Par definition, $Z/n(1)$ is the sheaf etale of roots (n)-iemes of l'unite.

(2.14) Indications bibliographiques on the SSSS 1 and 2.

A) Tous the results importants en cohomologie etale se demontrent d'abord for of sheaves of torsion. L'extension aux $\mathbb{Q}$ -sheaves se fait par of passages a the limite formels. Dans ce qui suit, for chaque theorem cite, je renverrai non point a a reference or il is demontre, mais a a reference or son analogue for the sheaves of torsion l'est.

B) A l'exception of the formule of Lefschetz and of (2.12), the results of cohomologie etale utilises in cet article are all demontres in SGA 4. For ceux already enonces, the references are : definition of H^i : VII ; definition of Hci : XVII 5.1 ; theorem of finitude : XIV 1, complete en XVII 5.3 ; dimension cohomologique : X ; dualite of Poincare : XVIII.

C) La relation entre the divers Frobenius ((1.4), (1.11), (1.15)) is expliquée en detail in SGA 5, XV, SSSS 1, 2.

D) L'interpretation cohomological of fonctions Z (1.14.3) is clairement exposee in [1] ; toutefois, the formule of Lefschetz (1.12), for X a courbe projective and smooth, y is utilisee, mais non demontree. For the proof, il faut, helas, renvoyer a SGA 5.

E) La form (2.12) of the theorem of Poincare duality resulte of the resultat general SGA 4, XVIII (3.2.4.1) (for $S = \text{Spec}(k)$, $X = X$, $K = j_! F$, $L = \mathbb{Q}$) par a calcul local qui n'est pas difficile. L'enonce figurera explicitement in the version definitive of SGA 5. Dans the case or we l'utiliserons (ramification of F moderee), one pourrait l'obtenir par voie transcendante, en relevant X and F en caracteristique 0.

3. The fundamental estimate

Le result of ce paragraphe a been catalyse par the lecture of Rankin [3].

(3.1) Let U_0 a courbe on F_q , complement in P^1 d'un ensemble fini of closed points, U the courbe on F_q qui s'en deduit, u a point ferme of U , F_0 a $\mathbb{Q}$ -sheaf constant tordu on U_0 and F son image reciproque on U .

Let $b \in \mathbb{Q}$. Nous dirons que F_0 is of weights b if for all $x \in |U_0|$, the eigenvalues of F_x agissant on F_0 (1.13) are of nombres algebriques dont all the conjuges complexes are of value absolue $b/2$.

qx . Par exemple, $\mathbb{Q}(r)$ is of weights $-2r$.

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Theoreme 3.1 (3.2). Faisons the hypotheses suivantes :

(i) F_0 is muni d'une form bilineaire alternee non degeneratese

$$\text{ps} : F_0 \otimes F_0 \rightarrow \mathbb{Q}(-b).$$

(ii) L'image of $n_1(U, u)$ in $GL(F_u)$ is a sous-groupe ouvert of the group symplectique $Sp(F_u, \text{psu})$.

(iii) For all $x \in |U_0|$, the polynome $\det(1 - F_x t, F_0)$ is a coefficients rationnels.

Then, F is of weights b .

On peut supposer, and we supposons, que U is affine and que $F \neq 0$.

Lemme 3.2 (3.3). Let $2k$ a entier pair and notons $F \otimes 2k$ the puissance tensorielle $(2k)$ -ieme of F_0 . For $x \in |U|$, the derivee logarithmique

$$\frac{d}{dt} \log(\det(1 - Fx \deg(x), F \otimes 2k)^{-1})$$

is a serie formelle a coefficients rationnels positifs.

L'hypothese (iii) assure que, for all n , $\text{Tr}(F^n, F_0) \in \mathbb{Q}$. Le nombre

$$\text{Tr}(F^n, F \otimes 2k) = \text{Tr}(F^n, F_0)^{2k}$$

is therefore rationnel positif, and one applique (??).

Lemme 3.3 (3.4). Les facteurs locaux $\det(1 - Fx \deg(x), F \otimes 2k)^{-1}$ are of series formelles a coefficients rationnels positifs.

La serie formelle $\log \det(1 - Fx \deg(x), F \otimes 2k)^{-1}$ is without terme constant ; after (??), ses coefficients are ≥ 0 ; the coefficients of son exponentielle are therefore egalement positifs.

P

Lemme

3.4 (3.5). Let $f = \sum_{n=0}^{\infty} a_n t^n$ the product of series formelles $f_i = \sum_{n=0}^{\infty} a_{i,n} t^n$, all of terme constant a , and a coefficients reels positifs.

On suppose

que l'ordre of $f_i - 1$ tend vers l'infini with i , and one pose

$f = \prod_i f_i$. Then, the rayon of convergence absolue of f_i is au less egal a celui of f .

P

If $f = \sum_{n=0}^{\infty} a_n t^n$, one a en effet $a_{i,n} \leq a_n$.

Lemme 3.5 (3.6). Sous the hypotheses of (??), if f and the f_i are the developpements en serie of Taylor of fonctions meromorphes, then

$$\inf\{|z| \mid f(z) = \infty\} \leq \inf\{|z| \mid f_i(z) = \infty\}.$$

Ces nombres are en effet the rayons of convergence absolue.

(3.7) For chaque partition P of $[1, 2k]$ en parties a deux elements $\{i_a, j_a\}$ ($i_a < j_a$), one defines

$$\text{ps}_P : F \otimes 2k \rightarrow Q \otimes (-kb) : x_1 \otimes \dots \otimes x_{2k} \mapsto \sum_a \text{ps}(x_{i_a}, x_{j_a}).$$

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Let x a point ferme of X . L'hypothese (ii) assure que the coinvariants of $\pi_1(U, u)$ in $F \otimes 2k$ are the coinvariants in $F \otimes 2k$ of the group symplectique all entier

$$\text{Sp}(F_u, \text{ps}_u) \subset \text{GL}(F_u).$$

(Le group symplectique agit on $F \otimes 2k$; l'ensemble of coinvariants is l'ensemble of invariants of the dual). After H. Weyl (The classical groups, Princeton University Press, chap. VI, SS 1), for $P \in P$ convenable, dependant of $\dim(F_u)$, the ps_P for $P \in P$ definissent a isomorphism

$$(F \otimes 2k)^{\text{Sp}} \rightarrow Q \otimes (-kb)^P.$$

Let N the nombre d'elements of P . After (??), the formule ci-dessus gives

$$\text{Hc}_{2k}(U, F \otimes 2k) = Q \otimes (-kb)^{\otimes N}.$$

Puisque $\text{Hc}_1(U, F \otimes 2k) = 0$, the formule (??) se reduit a

$$Z(U_0, F \otimes 2k, t) = \det(1 - F * t, \text{Hc}_{2k}(U, F \otimes 2k))^{-1}.$$

Cette fonction Z is therefore the developpement en serie of Taylor d'une fonction rationnelle n'ayant of pole qu'en $t = 1/q^{kb}$. Nous utiliserons seulement the fait que the poles are of value absolue $1/q^{kb}$ in $\mathbb{C}$. Cela pourrait se deduire d'arguments generaux on the groups reductifs. If a is a value proper of $F *$ on F_0 , then a^{2k} is a value proper of $F *$ on $F \otimes 2k$. Notons again a a quelconque conjugue complexe of a .

La puissance inverse $1/a^{2k \deg(x)}$ is a pole of $\det(1 - Fx \deg(x), F \otimes 2k)$. After (??) and (??), one a therefore

$$|1/a^{2k}| \leq 1/q^{kb}$$

let

$$|a| \geq q^{b/2}.$$

Par ailleurs, l'existence of ps assure que q^{b-a-1} is egalement value proper, d'ou l'inegalite

$$|q^{b-a-1}| \geq q^{b/2}$$

let

$$q^{b/2} \geq |a|.$$

Ceci acheve the proof.

Corollaire 3.6 (3.8). Let a a value proper of $F *$ agissant on $\text{Hc}_1(U, F)$. Then, a is a nombre algebrique, dont all the conjugues complexes satisfy

$$q^{(b+1)/2} \leq |a| \leq q^{(b+1)/2}.$$

La formule (??) for F_0 se réduit a
 $Z(U_0, F_0, t) = \det(1 - F * t, H^1(U, F))$.
 Le premier membre is a serie formelle a coefficients rationnels, vu
 son developpement en product and l'hypothese (iii). Le second membre

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is therefore a polynome a coefficients rationnels ; $1/a$ en is root. Ceci
 proves already que a is algebrique. For achever the proof, il
 suffices of verifier que the product infini qui defines $Z(U_0, F_0, t)$ converge
 absolument (therefore is non zero) for $|t| < q^{-(b+1)/2}$.
 Let N the rang of F , and posons

$$\det(1 - Fx t, F) = \prod_{i=1}^N (1 - a_{i,x} t).$$

After (??), $|a_{i,x}| = q^x$. La convergence of the product infini Z resulte
 of celle of

$$\prod_{x,n} |a_{i,x}|^n |t^{n \cdot \deg(x)}|.$$

For $|t| = q^{-(b+1)/2 - \epsilon}$ ($\epsilon > 0$), one a

$$\prod_{i,x} |a_{i,x}|^n |t^{n \cdot \deg(x)}| = N \prod_x q^{-n \cdot \deg(x)(1+\epsilon)}.$$

Sur the droite affine, il y a q^n points a values in F_q^n , therefore au more
 q^n closed points of degree n . On a therefore

$$\prod_n q^n * q^{-n(1+\epsilon)} = \prod_n q^{-n\epsilon} < \infty,$$

ce qui acheve the proof.

Corollaire 3.7 (3.9). Let j_0 l'inclusion of U_0 in $P_1 F_q$, j celle of
 U in P_1 , and a a value proper of F * agissant on $H^1(P_1, j_* F)$.
 Then a is a nombre algebrique, dont all the conjugues complexes
 satisfy

$$q^{(b+1)/2-1} \leq |a| \leq q^{(b+1)/2}.$$

Un segment of the sequence longue of cohomologie definie par the sequence
 exacte courte

$$0 \rightarrow j_! F \rightarrow j_* F \rightarrow j_* F / j_! F \rightarrow 0$$

($j_! =$ prolongement par 0) s'ecrit

$$H^1_c(U, F) \rightarrow H^1(P_1, j_! F) \rightarrow H^1(P_1, j_* F) \rightarrow 0.$$

La value proper a apparait therefore already in $H^1(U, F)$, and is justiciable
 de (??) :

$$q^{(b+1)/2-1} \leq |a| \leq q^{(b+1)/2}.$$

La Poincare duality (2.12) assure que $q^{b+1-a-1}$ is egalement value
 proper, d'ou l'inegalite

$$|q^{b+1-a-1}| \leq q^{(b+1)/2}$$

and the corollary.

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4. Lefschetz theory: local theory

(4.1) Sur C , the results locaux of Lefschetz are the suivants.

Let $D = \{z \mid |z| < 1\}$ the disque unite, $D^* = D - \{0\}$, and $f : X \rightarrow D$
 a morphism d'espaces analytiques. On suppose que :

- a) X is non singulier, and purement of dimension $n + 1$;
- b) f is proper ;
- c) f is smooth en dehors d'un point x of the fibre speciale $X_0 = f^{-1}(0)$;
- d) en x , f presente a point quadratique non degenerates.

Let $t \neq 0$ in D and $X_t = f^{-1}(t)$ "the" fibre generale. Aux given
 precedentes, one associe :

- a) of morphisms of specialisation $sp : H^i(X_0, Z) \rightarrow H^i(X_t, Z)$:
 X_0 is a retracte par deformation of X , and sp is the fleche com-
 $\sim$
 $\rightarrow H^i(X_0, Z) \leftarrow$
 $\sim$
 $\rightarrow H^i(X, Z) \rightarrow H^i(X_t, Z)$;
- b) of transformations of monodromie $T : H^i(X_t, Z) \rightarrow H^i(X_t, Z)$,
 qui decrivent l'effet on the cycles singuliers of X_t of "faire tourner
 t autour of 0". C'est again l'action on $H^i(X_t, Z)$, fibre en t of the
 systeme local $Ri f_* Z|D^*$, of the generateur positif of $\pi_1(D^*, t)$.

The theory of Lefschetz decrit a) and b) en terme of the cycle evanescent
 $d \in H^n(X_t, Z)$. Ce cycle is bien defini au signe pres. For $i \neq n, n + 1$,
 one a

$$H^i(X_0, Z) \xrightarrow{\sim} H^i(X_t, Z) \quad (i \neq n, n + 1).$$

For $i = n, n + 1$, one has a sequence exacte

$$0 \rightarrow H^n(X_0, Z) \rightarrow H^n(X_t, Z) \xrightarrow{x \mapsto x + d} H^{n+1}(X_0, Z) \rightarrow H^{n+1}(X_t, Z) \rightarrow 0.$$

For $i \neq n$, the monodromie T is l'identite. For $i = n$, one a

$$T x = x + (x, d)d.$$

Les values of (x, d) , of $T d$, and of (d, d) are the suivantes :

$n \bmod 4$	0	1	2	3
(x, d)	+	-	-	+
$T d$	2	0	-2	0
(d, d)	d	d	-d	-d

La transformation of monodromie T respecte the form d'intersection $\text{Tr}(x \cup y)$ on $H^n(X_t, Z)$. For n impair, c'est a transvection symplectique. For n pair, c'est a symetrie orthogonale.

(4.2) Voici l'analogie of (4.1) en geometrie algebrique abstraite. Le disque D is remplace par the spectre d'un anneau of valuation discrete henselien A a residue field algebraically closed. Let S ce spectre, e son point generique (spectre of the corps of fractions of A), s son point ferme (spectre of the residue field). Le role of t is joue par a point generique geometric e (spectre d'une cloture algebrique of the corps of fractions of A).

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Let $f : X \rightarrow S$ a morphism proper, with X regulier purement of dimension $n + 1$. On suppose f smooth, sauf for a point quadratique ordinaire x in the fibre speciale X_s .

Let ℓ a nombre premier different of the caracteristique residuelle p of S . Notant X_e the fibre generique geometric, one dispose again d'un morphism of specialisation

$$(4.1) \quad \text{sp} : H^i(X_s, \mathbb{Q}_\ell) \xrightarrow{\sim} H^i(X, \mathbb{Q}_\ell) \rightarrow H^i(X_e, \mathbb{Q}_\ell).$$

Le role of T is joue par l'action of the group d'inertie $I = \text{Gal}(e/e)$ agissant on $H^i(X_e, \mathbb{Q}_\ell)$ par transport of structure (cf. (1.15)) :

$$(4.2) \quad I = \text{Gal}(e/e) \rightarrow \text{GL}(H^i(X_e, \mathbb{Q}_\ell)).$$

Les given (??), (??) decrivent entierement the sheaves $R_i f_* \mathbb{Q}_\ell$ on S .

(4.3) Posons $n = 2m$ for n pair, and $n = 2m + 1$ for n impair. (??)

and (??) se decrivent again en terme d'un cycle evanescent

$$(4.3) \quad d \in H^n(X_e, \mathbb{Q}_\ell)(m).$$

Ce cycle is bien defini au signe pres.

For $i \neq n, n + 1$, one a

$$(4.4) \quad H^i(X_s, \mathbb{Q}_\ell) \xrightarrow{\sim} H^i(X_e, \mathbb{Q}_\ell) \quad (i \neq n, n + 1).$$

For $i = n, n + 1$, one a a sequence exacte

$$(4.5) \quad 0 \rightarrow H^n(X_s, \mathbb{Q}_\ell) \rightarrow H^n(X_e, \mathbb{Q}_\ell) \xrightarrow{x \mapsto x + d} \mathbb{Q}_\ell(m-n) \rightarrow H^{n+1}(X_s, \mathbb{Q}_\ell) \rightarrow H^{n+1}(X_e, \mathbb{Q}_\ell) \rightarrow 0.$$

L'action (??) of I (the monodromie locale) is triviale if $i \neq n$. For $i = n$, elle se decrit as suit.

A) n impair. -- On dispose d'un homomorphisme canonical

$$t : I \rightarrow \mathbb{Z}/(1),$$

and l'action of $s \in I$ is

$$sx = x + t(s)(x, d)d.$$

B) n pair. -- Ce case ne we servira pas. Disons seulement que, if $p \neq 2$, il existe a unique caractere d'ordre deux

$$e : I \rightarrow \{\pm 1\},$$

and que l'on a

$$sx = x \text{ si } e(s) = +1, \\ sx = x + (x, d)d \text{ if } e(s) = -1.$$

Les signes $\pm$ in A) and B) are the memes qu'en (4.1).

(4.4) Ces results apportent the informations suivantes on the $R_i f_* \mathbb{Q}_\ell$.

a) Si $d \neq 0$:

$6 \leq n$, the sheaf $R_i f_* \mathbb{Q}_\ell$ is constant.

1) Pour $i =$

2) Let j l'inclusion of e in S . On a $R^n f_* \mathbb{Q}_\ell = j_* j^* R^n f_* \mathbb{Q}_\ell$.

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b) If $d = 0$: (C'est the a case exceptionnel. Puisque $(d, d) = \pm 2$ for n pair, il ne peut se produire que for n impair.)

1) For $i \neq n + 1$, the sheaf $R_i f_* \mathbb{Q}_\ell$ is constant.

2) Let $\mathbb{Q}_\ell(m-n)$ the sheaf $\mathbb{Q}_\ell(m-n)$ on $\{s\}$, prolonge par zero on S . On a a sequence exacte

$0 \rightarrow Q \rightarrow (m-n)s \rightarrow R_{n+1} \rightarrow f_* Q \rightarrow j_* j^* R_{n+1} \rightarrow f_* Q \rightarrow 0$,
or $j_* j^* R_{n+1} \rightarrow f_* Q$ is a sheaf constant.

5. Lefschetz theory: global theory

(5.1) Sur C , the results of Lefschetz are the suivants. Let P a space projectif of dimension > 1 , and P^* l'espace projectif dual ; ses points parametrisent the hyperplans of P , and one note H_t l'hyperplan defini par $t \in P^*$. If A is a sous-espace lineaire of codimension 2 of P , the hyperplans contenant A are parametrises par the points d'une droite $D \subset P^*$, the duale of A . Ces hyperplans $(H_t)_{t \in D}$ forment the pinceau d'axe A .

Let $X \subset P$ a variety projective nonsingular connexe and of dimension $n+1$. Let $X \subset X \times D$ l'ensemble of couples (x, t) tels que $x \in H_t$. Les maps premiere and seconde coordonnee forment a diagramme

$$(5.1) \quad \begin{array}{ccc} X & & /X \\ f \downarrow & & \downarrow \\ D & \square & / \text{Spec}(C) \end{array}$$

La fibre of f en $t \in D$ is the section hyperplane $X_t = X \cap H_t$ of X .

Fixons X , and prenons A assez general. Then :

A) A is transverse a X , and X se deduit of X par eclatement of $A \cap X$. En particulier, X is non singulier.

B) Il existe a partie finie S of D and for chaque $s \in S$ a point $x_s \in X_s$, tels que f let smooth en dehors of x_s .

C) Les x_s are of points critiques non degenerates of f .

For chaque $s \in S$, the theorie of Lefschetz locale (4.1) s'applique therefore a a petit disque D_s autour of s and a $f^{-1}(D_s)$.

(5.2) On pose $U = D - S$. Let $u \in U$, and choisissons of lacets disjoints

$(g_s)_{s \in S}$ partant of u , with g_s tournant a fois autour of s :

Ces lacets engendrent the fundamental group $\pi_1(U, u)$. Ce group agit on $H^i(X_u, \mathbb{Z})$, fibre en u of the systeme local $R_i \otimes \mathbb{Z}[U]$. After the theorie locale (4.1), a chaque $s \in S$ corresponds a cycle evanescent $ds \in H^n(X_u, \mathbb{Z})$; ces cycles dependent of the choix of g_s . For $i \neq n$, l'action of $\pi_1(U, u)$ on $H^i(X_u, \mathbb{Z})$ is triviale. For $i = n$, one a

$$(5.2) \quad g_s x = x + (-1)^n (x, ds) ds.$$

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Let E the sous-espace of $H^n(X_u, \mathbb{Q})$ engendre par the ds (partie evanescente of the cohomology).

Proposition 5.1 (5.3). E is stable sous l'action of the group of monodromie $\pi_1(U, u)$. L'orthogonal $E^\perp$ of E (for the form d'intersection $\text{Tr}(x \cup y)$) is the sous-espace of invariants of the monodromie in $H^n(X_u, \mathbb{Q})$.

Les g_s engendrant the group of monodromie, c'est clair on (??).

Theoreme 5.2 (5.4). Les cycles evanescents $\pm ds$ (pris au signe pres) are conjugues sous l'action of $\pi_1(U, u)$.

Let $X \subset P$ the variety duale of X : c'est l'ensemble of $t \in P^*$ tels que H_t let tangent a X , i.e. tels que X_t let singulier, or que $X \subset H_t$. La variety X is irreductible. Let $Y \subset X \times P^*$ l'espace of couples (x, t) tels que $x \in H_t$. On dispose d'un diagramme

$$\begin{array}{ccc} Y & & /X \\ g \downarrow & & \downarrow \\ P & \square & / \text{Spec}(C) \end{array}$$

La fibre of g en $t \in P^*$ is the section hyperplane $X_t = X \cap H_t$ of X , and g is smooth en dehors of l'image reciproque of X .

On retrouve the situation of (??) en remplaçant P par a droite $D \subset P$ and Y par $g^{-1}(D)$. On a $S = D \cap X$.

After a theorem of Lefschetz, for D assez generale,

$$\pi_1(D - S, u) \rightarrow \pi_1(P - X, u)$$

is surjective. Il suffices therefore of show que the $\pm ds$ are conjugues sous $\pi_1(P - X)$.

For x in the lieu smooth of codimension 1 of X , let ch a chemin of u a x in $P - X$, and g_x the lacet qui suit ch jusqu'au voisinage of X , tourne a fois autour of X , puis revient a u par ch . Les lacets g_x (for ch variable) are conjugues entre eux. Puisque X is irreductible, deux points of the lieu smooth of X peuvent always, in X , be joints par a chemin qui ne quitte pas the lieu smooth. Il en resulte que the classe of conjugaison of g_x ne depend pas of x . En particulier, the g_s are conjugues entre eux. On lit on (??) que this implique the conjugaison des $\pm ds$.

Corollaire 5.3 (5.5). L'action of $\pi_1(U, u)$ on $E/(E \cap E^\perp)$ is absolument irreductible.

Let $F \subset E/(E \cap E)$ a sous-espace stable sous the monodromie. If $F \not\subset (E \cap E)$, il existe $x \in F$ and $s \in S$ tels que $(x, ds) \neq 0$. On a then $gs - x = \pm(x, ds)ds \in F$,

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and $\pm ds \in F$. After (??), all the $\pm ds$ are then in F , and $F = E$. Ceci proves (??).

(5.6) Ces results se transposent as suit en geometrie algebrique abstraite.

Let P a space projectif of dimension ≥ 1 on k algebriquement clos of caracteristique p and $X \subset P$ a variety projective nonsingular connexe and of dimension $n + 1$.

For A a sous-espace lineaire of P of codimension 2, one defines as en (5.1) D and the pinceau $(H_t)_{t \in D}$, the X_t , X and the diagramme (??). On dit que the $(X_t)_{t \in D}$ forment a pinceau of Lefschetz of sections hyperplanes if the conditions suivantes are verifiees :

A) L'axe A is transverse a X . L'espace X se deduit then of X par eclatement of $A \cap X$, and is smooth.

B) Il existe a partie finie S of D and for chaque $s \in S$ a point $x_s \in X_s$, tels que f let smooth en dehors of x_s .

C) x_s is a point singulier quadratique ordinaire of X_s .

For chaque $s \in S$, the theorie of Lefschetz locale of the SS_4 s'applique au spectre D_s of l'henselise of l'anneau local of D en s , and a $X_s = X \times_D D_s$.

(5.7) Let N the dimension of P , r a entier ≥ 1 , $\square$ and $i(r)$ the plongement of P in l'espace projectif of dimension $N+r-1$, of coordonnees homogenes the monomes of degre r en the coordonnees homogenes of P . Les sections hyperplanes of $i(r)(P)$ are the hypersurfaces of degre r of P .

If $p \neq 0$, il se peut qu'aucun pinceau of sections hyperplanes of X ne let of Lefschetz. Toutefois, if $r \geq 2$ and qu'on remplace the plongement projectif gives $i : X \rightarrow P$ par $i_0 = i(r) * i$, then, in ce nouveau plongement, all pinceau assez general is of Lefschetz. En d'autres termes, if $r \geq 2$, a pinceau assez general of sections hypersurfaces of degre r of X is always of Lefschetz.

(5.8) Dans the sequence of cette discussion, we etudions a pinceau of Lefschetz of sections hyperplanes of X , en excluant the case $p = 2$, n pair. Le case or n is impair we suffirait for the sequence. On pose $U = D - S$. Let $u \in U$ and ℓ a nombre premier $\neq p$. Les results locaux of the SS_4 montrent que $R_n f^* \mathbb{Q}_\ell$ is moderelement ramifie en chaque $s \in S$.

Le fundamental group modere of U is a quotient of the complete profini of the fundamental group transcendant analogue (relevement en caracteristique 0 of revetements moderes, and theorem d'existence of Riemann). La situation algebrique is of lors all pareille a the if-tuation transcendant, and the transposition of results of Lefschetz se fait par of arguments standards. Dans the proof of (??), the theorem of Lefschetz on the n_1 devient the theorem of Bertini, and one doit invoquer the lemma d'Abhyankar for controler the ramification of $R_i g^* \mathbb{Q}_\ell$ the long of the lieu smooth of codimension a of X .

Les results are the suivants.

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a) If the cycles evanescents are non zero :

$\neq 0$, the sheaf $R_i f^* \mathbb{Q}_\ell$ on D is constant.

1) Pour $i =$

2) Let j l'inclusion of U in D . On a $R_n f^* \mathbb{Q}_\ell = j^* j_* R_n f^* \mathbb{Q}_\ell$.

3) Let $E \subset H^n(X_u, \mathbb{Q}_\ell)$ the sous-espace of the cohomology engendre par the cycles evanescents. Ce sous-espace is stable sous $\pi_1(U, u)$, and $E = H^n(X_u, \mathbb{Q}_\ell) \cap \pi_1(U, u)$. La representation of $\pi_1(U, u)$ on $E/(E \cap E)$ is absolument irreductible, and l'image of π_1 in $GL(E/(E \cap E))$ is engendree (topologiquement) par the $x \mapsto x \pm (x, ds)ds$ ($s \in S$) (the signe $\pm$ being determine as en (4.1)).

b) If the cycles evanescents are zero : (C'est the a case exceptionnel.

Puisque $(d, d) = \pm 2$ for n pair, il ne peut se produire que for n impair : $n = 2m + 1$. On notera que if a cycle evanescent is zero, ils the are all, car conjugues.)

1) For $i \neq n + 1$, the sheaf $R_i f^* \mathbb{Q}_\ell$ is constant.

2) On a a sequence exacte

$$0 \rightarrow \bigoplus_{s \in S} \mathbb{Q}_\ell(m - n)_s \rightarrow R_{n+1} f^* \mathbb{Q}_\ell \rightarrow F \rightarrow 0,$$

or F is constant.

3) $E = 0$.

(5.9) Le sous-espace $E \cap E$ of E is the kernel of the restriction a E of the form d'intersection $Tr(x \cup y)$. Cette form induit therefore a form bilineaire non degeneratese

$\rho : E/(E \otimes E) \otimes E/(E \otimes E) \rightarrow Q(-n)$,
 alternee for n impair, and symetrique for n pair. Cette form is
 respectee par the monodromie ; for n impair, the representation of mo-
 nodromie induit therefore

$$r : \pi_1(U, u) \rightarrow \text{Sp}(E/(E \otimes E), \rho).$$

Theoreme 5.4 (5.10). (Kajdan-Margulis). L'image of r is ouverte.

L'image of r is a sous-groupe compact, therefore analytique $\mathbb{A}$ -adique,
 of $\text{Sp}(E/(E \otimes E), \rho)$. Il suffices of show que son algebre of Lie
 L is egale a $\text{sp}(E/(E \otimes E), \rho)$. L'analogue transcendant of cette
 algebre of Lie is l'algebre of Lie of l'adherence of Zariski of the group
 of monodromie.

On deduit of (5.8) que L is engendree par the transformations of
 carre zero

$$N_s : x \mapsto x + (x, ds)ds \quad (s \in S)$$

and que $E/(E \otimes E)$ is a representation absolument irreductible of
 L . Le theorem resulte of the lemma suivant.

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Lemme 5.5 (5.11). Let V a space vectoriel of dimension finie
 on a corps k of caracteristique 0 , ρ a form alternee non degeneratese
 and L a sous-algebre of Lie of $\text{sp}(V, \rho)$. On suppose que :

- (i) V is a representation simple of L .
- (ii) L is engendree par a famille d'endomorphisms of V of the
 form $x \mapsto x + \rho(x, d)d$.

Then, $L = \text{sp}(V, \rho)$.

On peut supposer and one supposera que V , therefore ρ , is non zero. Let
 $W \subset V$ l'ensemble of $d \in V$ tels que $N(d) : x \mapsto x + \rho(x, d)d$ let in
 L .

- a) W is stable par homothetie (car L is a sous-espace vectoriel
 de $\text{gl}(V)$).
- b) If $d \in W$, $\exp(tN(d))$ is a automorphisme of (V, ρ, ρ) , therefore
 transforme W en lui-meme. If $d \neq 0$, $d \in W$, one a therefore $\exp(tN(d)) \cdot d \neq 0$
 $= d + t\rho(d, d)d \in W$; if $\rho(d, d) \neq 0$, the sous-espace vectoriel
 tendu par d and d is in W .
- c) Il en resulte que W is reunion of ses sous-espaces lineaires maxi-
 maux W_α , and que ceux-ci are deux a deux orthogonaux. Chaque
 W_α is of lors stable sous the $N(d)$ ($d \in W$), therefore stable sous
 L . Vu l'hypothese (i), $W \neq 0$ and L contient all the $N(d)$ for
 $d \in V$. On conclut en notant que l'algebre of Lie $\text{sp}(V, \rho)$ is
 engendree par the $N(d)$ ($d \in V$).

Remarque 5.6 (5.12). (inutile for the sequence). Il is maintenant facile
 of prouver (??) for a hypersurface of dimension impaire n in
 $\mathbb{P}^{n+1}$

F_q .

Let X_0 a telle hypersurface, and X l'hypersurface on F_q qui s'en
 deduit par extension of scalaires. On a

$$H^{2i}(X, Q) = Q(-i) \quad (0 \leq i \leq n);$$

2i

$H(X, Q(i))$ is engendre par the ieme cup-puissance of x , the classe of
 cohomologie $c_1(0(1))$ d'une section hyperplane. On a therefore

$$Z(X_0, t) = \det(1 - F_t, H(X, Q)) \cdot \prod_{i=0}^n (1 - q^i t)^{-1}$$

and $\det(1 - F_t, H(X, Q))$ is a polynome a coefficients entiers inde-
 n

pendants of $\mathbb{A}$.

Faisons varier X_0 in a pinceau of Lefschetz d'hypersurfaces, qui
 let defini on F_q (cf. (5.7) for $X = \mathbb{P}^{n+1}$; l'existence d'un tel pin-
 ceau n'est pas claire ; if one voulait completer l'argument esquisse ici, il
 faudrait recourir aux arguments qui will be donnees en (7.1)). On satisfies
 que E coincide ici with the H^n all entier, and (??) fournit the conjecture
 of Weil for all the hypersurfaces of the pinceau, en particulier for
 X_0 .

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(5.13) Indications bibliographiques on the SSSS 4 and 5.

A) Les results of Lefschetz (4.1) and (5.1) a (5.5) are contenus in
 son livre [2]. For the theorie locale (4.1), il peut be more commode of
 consulter SGA 7, XIV (3.2).

B) Les results of the SS 4 are demontres in the exposes XIII, XIV and

XV of SGA 7.

C) (5.7) is demontre in SGA 7, XVII.

D) (5.8) is demontre in SGA 7, XVIII. Le theoreme d'irreductibilite y is demontre for E, mais seulement sous l'hypothese que $E \cap E^\perp = \{0\}$. La proof in the case general (for $E/(E \cap E^\perp)$) is pareille.

6. A rationality theorem

(6.1) Let P_0 a space projectif of dimension > 1 on F_q , $X_0 \subset P_0$ a variety projective nonsingular, $A_0 \subset P_0$ a sous-espace lineaire of codimension deux, $D_0 \subset P_0$ the droite duale, F_q a cloture algebrique of F_q and P , X , A , D on F_q deduits of P_0 , X_0 , A_0 , D_0 par extension of scalaires. Le diagramme (??) of (5.6) provient d'un diagramme analogue on F_q :

$$(6.1) \quad \begin{array}{ccc} X_0 & & / X_0 \\ f_0 & & \\ \square & & \square \\ D_0 & & / \text{Spec}(F_q) \end{array}$$

On suppose que X is connexe of dimension paire $n + 1 = 2m + 2$, and que the pinceau $(X_t)_{t \in D}$ of sections hyperplanes of X defini par D is a pinceau of Lefschetz.

L'ensemble S of $t \in D$ tels que X_t let singulier is defini on F_q , i.e. provient of $S_0 \subset D_0$. On pose $U_0 = D_0 - S_0$ and $U = D - S$.

Let $u \in U$. La partie evanescente of the cohomology $E \subset H^n(X_u, \mathbb{Q})$ is stable sous $\pi_1(U, u)$, therefore defines on U a sous-systeme local E of $R_n f_* \mathbb{Q}$. Ce dernier is defini on F_q : $R_n f_* \mathbb{Q}$ is l'image reciproque of the $\mathbb{Q}$ -sheaf $R_n f_0^* \mathbb{Q}$ on D_0 , and, on U , E is l'image reciproque d'un sous-systeme local $E_0 \subset R_n f_0^* \mathbb{Q}$.

Le cup-produit is a form alternee

$$ps : R_n f_0^* \mathbb{Q} \otimes R_n f_0^* \mathbb{Q} \rightarrow \mathbb{Q}(-n).$$

Notant E_0 l'orthogonal of E_0 relativement a ps , in $R_n f_0^* \mathbb{Q} |_{U_0}$, one voit que ps induit a dualite parfaite

$$ps : E_0 / (E_0 \cap E_0^\perp) \otimes E_0 / (E_0 \cap E_0^\perp) \rightarrow \mathbb{Q}(-n).$$

Theoreme 6.1 (6.2). For all $x \in |U_0|$, the polynome $\det(1 - F_x^* t, E_0 / (E_0 \cap E_0^\perp))$ is a coefficients rationnels.

Corollaire 6.2 (6.3). Let j_0 l'inclusion of U_0 in D_0 , and j celle of U in D . Les eigenvalues of F^* agissant on $H^1(D, j_* E_0 / (E_0 \cap E_0^\perp))$ are of nombres algebriques dont all the conjugues complexes a satisfy

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E_0)) are of nombres algebriques dont all the conjugues complexes a satisfy

$$q^{(n+1)/2-1} \leq |a| \leq q^{(n+1)/2}.$$

After (??) and (??), the hypotheses of (??) are en effet verifiees par $(U_0, E_0 / (E_0 \cap E_0^\perp), ps)$, for $b = n$, and one applique (??).

Lemme 6.3 (6.4). Let G_0 a $\mathbb{Q}$ -sheaf constant tordu on U_0 , tel que son image reciproque G on U let a sheaf constant. Il existe then in $\mathbb{Q}$ of unites a_i telles que, for all $x \in |U_0|$, one ait

$$\det(1 - F_x^* t, G_0) = \prod_i^{\deg(x)} (1 - a_i t).$$

Ce lemma exprime que G_0 is l'image reciproque d'un sheaf on $\text{Spec}(F_q)$, a savoir son image directe on $\text{Spec}(F_q)$. Ce dernier s'identifie a a representation ℓ -adique G_0 of $\text{Gal}(F_q/F_q)$, and one prend

$$\det(1 - F t, G_0) = \prod_i^Y (1 - a_i t).$$

Le lemma (??) s'applique aux $R_n f_0^* \mathbb{Q}$ ($i \leq n$), a $R_n f_0^* \mathbb{Q} / E_0$ and a

$E_0 \cap E_0^\perp$.

For $x \in |U_0|$, the fibre $X_x = f_0^{-1}(x)$ is a variety on the finite field $k(x)$. If x is a point of U au-dessus of x , X_x se deduit of X_x par extension of scalaires of $k(x)$ a sa cloture algebrique $k(x) = F_q$, and $H^i(X_x, \mathbb{Q})$ is the fibre of $R_i f_* \mathbb{Q}$ en x . La formule (??) for the variety X_x on $k(x)$ s'ecrit therefore

$$Z(X_x, t) = \prod_i^Y \det(1 - F_x^* t, R_i f_0^* \mathbb{Q})^{(-1)^{i+1}}$$

and $Z(X_x, t)$ is the product of

Y

$$Z_1 = \det(1 - F_x * t, R_n f * Q / E_0) * \det(1 - F_x * t, E_0 n E_0) * \det(1 - F_x * t, R_i f_0 * Q) (-1)^{i+1}$$

i+1

$$i_6 = n$$

par

$$Z_m = \det(1 - F_x * t, E_0 / (E_0 n E_0)).$$

Posons $E_0 = E_0 / (E_0 n E_0)$, $E_e = E / (E n E)$ and appliquons (??) aux facteurs of Z_1 . On trouve qu'il existe of unites $\mathbb{A}$ -adiques a_i ($1 \leq i \leq N$) and b_j ($1 \leq j \leq M$) in Q tels que for all $x \in |U_0|$

$$Z(Xx, t) = i \prod_{j=1}^M (1 - a_i t)^{\deg(x)} * \prod_{j=1}^M (1 - b_j t)^{\deg(x)} \det(1 - F_x t, E_0)^j$$

and qu'en particulier the second membre is in $Q(t)$. If a a_i coincide with a b_j , il is loisible of supprimer simultanement cet a_i of the liste of a and ce b_j of the liste of b. On peut therefore supposer, and we supposerons, que $a_i \neq b_j$ for all i and all j.

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(6.5) Il suffices of prouver que the polynomes $i \prod_{j=1}^M (1 - a_i t)$ and $j \prod_{j=1}^M (1 - b_j t)$ are a coefficients rationnels, i.e. que the famille of a_i (resp. the famille of b_j) is definie on Q . Nous the deduirons of propositions suivantes. Proposition 6.4 (6.6). Let (g_i) ($1 \leq i \leq P$) and (d_j) ($1 \leq j \leq Q$) deux familles d'unites $\mathbb{A}$ -adiques in Q . On suppose que $g_i \neq d_j$. If K is a ensemble fini assez grand d'entiers $n \geq 1$, and L a partie of densite θ assez grande of $|U_0|$, then, if $x \in |U_0|$ satisfies $k \deg(x)$ (for all $k \in K$) and $x \in L$, the denominateur of

$$\det(1 - F_x * t, E_0)^{-1} * \prod_{i=1}^P (1 - g_i t)^{\deg(x)} \prod_{j=1}^Q (1 - d_j t)^{\deg(x)}$$

(6.2)

ecrit in the form irreductible, is $j \prod_{j=1}^Q (1 - d_j t)^{\deg(x)}$.

La proof will be given en (6.10-13). After (??) ci-dessous, (??) fournit a description intrinseque of the famille of d_j en terme of the famille of fractions rationnelles (??) for $x \in |U_0|$.

Lemme 6.5 (6.7). Let K a ensemble fini d'entiers $n \geq 1$ and (d_j) ($1 \leq j \leq Q$) and (e_j) ($1 \leq j \leq Q$) deux familles d'elements d'un corps. If, for all n assez grand, non divisible par no of $k \in K$, the famille of d_{jn} coincide with celle of e_{jn} (a l'ordre pres), then the famille of d_j coincide with celle of e_j (a l'ordre pres).

On procede par recurrence on Q . L'ensemble of entiers n tels que $dQ = e_{jn}$ is a ideal (n_j) . Prouvons qu'il existe J_0 tel que $dQ = e_{J_0}$. Sinon the n_j seraient distincts of 1, and il existerait of entiers arbitrairement grands n , non divisibles par no of n_j , ni par no of $k \in K$. On aurait $dQ = e_{jn}$ for all j , and this contredirait l'hypothese.

Il existe therefore J_0 tel que $dQ = e_{J_0}$. On conclut en appliquant l'hypothese of recurrence aux familles (d_j) ($j \neq J_0$) and (e_j) ($j \neq J_0$).

Proposition 6.6 (6.8). Let (g_i) ($1 \leq i \leq P$) Q and (d_j) ($1 \leq j \leq Q$) deux familles d'unites $\mathbb{A}$ -adiques in Q , $R(t) = i \prod_{i=1}^P (1 - g_i t)$ and $S(t) =$

$$j \prod_{j=1}^Q (1 - d_j t). \text{ Supposons que for all } x \in |U_0|, \prod_{i=1}^P (1 - g_i t)^{\deg(x)} \prod_{j=1}^Q (1 - d_j t)^{\deg(x)} \text{ divide } \det(1 - F_x * t, E_0)^{-1} * \prod_{i=1}^P (1 - g_i t)^{\deg(x)}.$$

Then $S(t)$ divide $R(t)$.

Supprimons of familles (g_i) and (d_j) of paires d'elements communs jusqu'a verifier l'hypothese of (??). Appliquons (??). Par hypothese, the fractions rationnelles (??) are of polynomes. Aucun d n'a therefore subsiste, ce qui signifie que $S(t)$ divide $R(t)$.

Cette proposition fournit a caracterisation intrinseque of $R(t)$ en

$$Q \prod_{i=1}^P (1 - g_i t)^{\deg(x)} * \det(1 - F_x * t, E_0).$$

terme of the famille of polynomes $Q \prod_{i=1}^P (1 - g_i t)^{\deg(x)}$. C'est the ppcm of polynomes $S(t) = j \prod_{j=1}^Q (1 - d_j t)$ verifiant l'hypothese

(6.9) Prouvons (6.5) and therefore (??) (modulo (??)). Faisons in (??) $(g_i) = (a_i)$ and $(d_j) = (b_j)$. On trouve a caracterisation intrinseque of the famille of b_j en terme of the famille of fractions rationnelles $Z(X_x, t)$ ($x \in |U_0|$). Celles-ci being in $Q(t)$, the famille of b_j is definie on Q .

$Q \quad \deg(x)$

Les polynomes $j(1 - b_j t)$ are therefore in $Q[t]$. La proposition (??) fournit a description intrinseque of the famille of a_i en terme of cette famille of polynomes. La famille of a_i is therefore definie on Q .

(6.10) Preliminaires. -- Let $u \in U$ and E_u the fibre of E en u . Le fundamental group arithmetique $\pi_1(U_0, u)$, extension of $Z = \text{Gal}(F_q/F_q)$ (generateur : ph) par the fundamental group geometric $\pi_1(U, u)$, agit on E_u par similitudes symplectiques :

$$r : \pi_1(U_0, u) \rightarrow \text{CSp}(E_u, \text{psu}).$$

Nous notons $\mu(g)$ the multiplicateur d'une similitude symplectique g . Let

$$H \subset Z^* \times \text{CSp}(E_u, \text{psu})$$

the sous-groupe defini par l'equation

$$q^{-n} = \mu(g)$$

(q being a unite ℓ -adique, $q \in Z^*$ is defini for all $n \in \mathbb{Z}$). Le fait que ps let a values in Q_ℓ^* ($-n$) s'exprime en disant que l'application of π_1 in $Z^* \times \text{CSp}$, of coordonnees the projection canonical on Z and r , se factorise par

$$r_1 : \pi_1(U_0, u) \rightarrow H.$$

Lemme 6.7 (6.11). L'image H_1 of r_1 is ouverte in H .

En effet, $\pi_1(U_0, u)$ se projette on Z , and l'image of $\pi_1(U, u) = \text{Ker}(\pi_1(U_0, u) \rightarrow Z)$ in $\text{Sp}(E_u, \text{psu}) = \text{Ker}(H \rightarrow Z^*)$ is ouverte (5.10).

Lemme 6.8 (6.12). For $z \in Q_\ell^*$ a unite ℓ -adique, l'ensemble Z of $(n, g) \in H_1$ tels que z n let value proper of g is a ferme of mesure zero.

Il is clair que Z is ferme. For chaque $n \in \mathbb{Z}$, let CSp_n l'ensemble of $g \in \text{CSp}(E_u, \text{psu})$ tels que $\mu(g) = q^{-n}$, and let Z_n l'ensemble of $g \in \text{CSp}_n$ tels que z n let value proper of g . Then, CSp_n is a space homogene sous Sp , and one satisfies que Z_n en is a sous-espace algebrique proper, therefore of mesure 0. After (??), $H_1 \cap (\{n\} \times Z_n)$ is therefore of mesure 0 in l'image inverse in H_1 of n , and one applique Fubini a the projection $H_1 \rightarrow Z$.

(6.13) Prouvons (??). For chaque i and j , l'ensemble of entiers n tels que $g_{in} = d_{jn}$ is l'ensemble of multiples d'un entier fixe n_0 (one n'exclut pas $n_0 = 0$). Par hypothese, $n_0 \neq 1$.

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After (??) and the theorem of densite of Cebotarev, l'ensemble of let value proper of F_{x^*} , agissant on E_{e_0} is

$$\deg(x)$$

$x \in |U_0|$ tels qu'un g_i of densite 0. On prend for K l'ensemble of n_0 and for L l'ensemble of x ci-dessus.

7. End of the proof of (1.7)

Lemme 7.1 (7.1). Let X_0 a variety projective nonsingular absolutely irreducible of dimension paire d on F_q . Let X on F_q deduit of X_0 par extension of scalaires and a a value proper of F^* agissant on $H_d(X, Q_\ell)$. Then, a is a nombre algebrique dont all the conjugues complexes, again notes a , satisfy

$$(7.1) \quad q^{d/2-1} \leq |a| \leq q^{d/2+1}.$$

On procede par recurrence on d (always suppose pair). Le case $d = 0$ is trivial, meme without supposer X_0 absolument irreducible ; one suppose desormais $d \geq 2$. On pose $d = n + 1 = 2m + 2$.

If Fr is a extension of degre r of F_q and que $X_0 \otimes Fr$ se deduit of X_0/F_q par extension of scalaires, l'assertion (??) for X_0/F_q equivaut a (??) for $X_0 \otimes Fr/F_r$: of meme que q is replace par q^r , the eigenvalues of F^* are remplacees par leurs puissances r -iemes.

After (5.7), in a plongement projectif convenable $i : X \rightarrow P$, X admet a pinceau of Lefschetz of sections hyperplanes. La remark precedente we permet of supposer i and ce pinceau definis on F_q (quitte a remplacer F_q par a extension finie).

Supposons therefore qu'il existe on F_q a plongement projectif $X_0 \rightarrow P_0$ and a sous-espace of codimension deux A_0 of P_0 qui definisse a pinceau of Lefschetz. On reprend the notations of (6.1) and (6.3). Une nouvelle extension of scalaires we ramene a supposer que :

a) Les points of S are definis on F_q .

- b) Les cycles evanescents en x_s ($s \in S$) are definis on F_q (puisque seul $+d$ is intrinseque, ils pourraient n'etre definis que on a extension quadratique).
- c) Il existe a point rationnel $u_0 \in U_0$. On prend the point correspondant u of U as point base.
- d) $Xu_0 = f_0^{-1}(u_0)$ admet a section hyperplane smooth Y_0 , definie on F_q . On pose $Y = Y_0 \otimes F_q$.

Puisque X se deduit of X par eclatement of the sous-variety smooth of codimension deux $A \cap X$, one a

$$H^i(X, Q) \sim H^i(X, Q)$$

(en fait, $H^i(X, Q) \sim H^i(X, Q) \otimes H^{i-2}(A \cap X, Q)(-1)$). Il suffices therefore of prouver (??) for the eigenvalues a of F * agissant on $H^d(X, Q)$.

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La spectral sequence of Leray for f s'ecrit

$$E_{p,q}^2 = H^p(D, R^q f_* Q) \Rightarrow H^{p+q}(X, Q).$$

Il suffices of prouver (??) for the eigenvalues a of F * agissant on

$E_{p,q}^\infty$ for $p + q = d = n + 1$.

Ce are :

$$1, n-1$$

A) $E_{1, n-1}^\infty$. After (5.8), $R^{n-1} f_* Q$ is constant. After (??), one a therefore

$$E_{1, n-1}^\infty = H^{n-1}(Xu, Q)(-1),$$

After the theorem of Lefschetz faible (corollary of SGA 4, XIV (3.2), and of the Poincare duality, SGA 4, XVIII), one a

$$H^{n-1}(Xu, Q)(-1) \xrightarrow{\sim} H^{n-1}(Y, Q)(-1),$$

and one applique l'hypothese of recurrence a Y_0 .

$$0, n+1$$

B) $E_{0, n+1}^\infty$.

If the cycles evanescents are non zero, $R^{n+1} f_* Q$ is constant and

$$0, n+1$$

$$E_{0, n+1}^\infty = H^{n+1}(Xu, Q).$$

Le morphism of Gysin

$$H^{n-1}(Y, Q)(-1) \rightarrow H^{n+1}(Xu, Q)$$

is surjectif (cet argument is dual of A)), and one applique l'hypothese of recurrence a Y_0 .

If the cycles evanescents are zero, the sequence exacte of (5.8) b) fournit a sequence exacte

$$\begin{aligned} M \\ Q(m-n) \rightarrow E_{0, n+1}^\infty \rightarrow H^{n+1}(Xu, Q). \end{aligned}$$

$$s \in S$$

La value proper of F agissant on $Q(m-n)$ is $q^{d/2}$, and H^{n+1} se traite as more haut.

$$2, n$$

C) $E_{2, n}^\infty$. If one disposait of the theorem of Lefschetz "difficile", one saurait que $E_{0, n}^\infty$ is zero, and que $R^n f_* Q$ is sum directe of $j^* E$ and d'un sheaf constant. Le H^1 d'un sheaf constant on P^1 is zero, and il suffirait d'appliquer (??).

Faute d'avoir already demontre Lefschetz "difficile", il we va falloir deviser. If the cycles evanescents are zero, $R^n f_* Q$ is constant ((5.8)

$$2, n$$

b)) and $E_{2, n}^\infty = 0$. On peut therefore supposer, and one supposera, the cycles evanescents non zero. Filtrons $R^n f_* Q = \sum j^* E$ (5.8) par the sous-sheaves $j^* E$ and $j^*(E + E)$.

If the cycles evanescents d ne are pas in $E \cap E$, one dispose of sequences exactes :

$$(7.2) \quad 0 \rightarrow j^* E \rightarrow R^n f_* Q \rightarrow \text{sheaf constant} \rightarrow 0,$$

$$(7.3) \quad 0 \rightarrow \text{sheaf constant} \rightarrow j^* E / (E \cap E) \rightarrow j^* E \rightarrow 0.$$

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If, a Dieu ne plaise, the d are in $E \cap E$, one a $E \subset E$ and of sequences exactes :

$$(7.4)$$

$$0 \rightarrow \text{the sheaf constant } j^* E \rightarrow R^n f_* Q \rightarrow \text{a sheaf } F \rightarrow 0,$$

$$(7.5)$$

$$0 \rightarrow F \rightarrow \text{the sheaf constant } j^* j^* F \rightarrow Q(n-m)s \rightarrow 0.$$

Dans the premier case, the sequences exactes longues of cohomologie four-nissent

(7.1.20) $H^1(D, j^* E) \rightarrow H^1(R^n f_* Q) \rightarrow 0$,
 (7.1.30) $0 \rightarrow H^1(D, j^* E) \rightarrow H^1(D, j^* (E/(E \otimes E)))$,
 and one applique (??).

Dans the second case, elles fournissent

(7.1.40) $0 \rightarrow H^1(D, R^n f_* Q) \rightarrow H^1(D; F)$,
 M
 (7.1.50) $Q(n-m) \rightarrow H^1(D, F) \rightarrow 0$,
 $s \in S$

and one remark que F agit on $Q(n-m)$ par multiplication par $q^{d/2}$.

Lemme 7.2 (7.2). Let X_0 a variety projective nonsingular ab-solument irreductible of dimension d on F_q . Let X on F_q deduit of X_0 par extension of scalaires and a a value proper of F^* agis-sant on $H^d(X, Q)$. Then, a is a nombre algebrique dont all the conjugues complexes, again notes a , satisfy $|a| = q^{d/2}$.

Prouvons already que (??) $\Rightarrow$ (??). For X_0 projective nonsingular on F_q and i a entier, il faut prouver l'assertion suivante :

$W(X_0, i)$. Let X deduit of X_0 par extension of scalaires of F_q a F_q . If a is a value proper of F^* agissant on $H^i(X, Q)$, then a is a nombre algebrique dont all the conjugues complexes, again notes a , satisfy $|a| = q^{i/2}$.

- If F is an extension of degree n of F_q , and que $X_0 \otimes F$ se deduit of X_0 / F_q par extension of scalaires, then $W(X_0, i)$ is equivalent a $W(X_0 \otimes F, i)$: etendre the scalaires revient a remplacer a par an and q par q^n .
- If X_0 is purement of dimension n , $W(X_0, i)$ is equivalent a $W(X_0, 2n-i)$: this resulte of the Poincare duality.
- If X_0 is sum of varieties X_{0a} , $W(X_0, i)$ is equivalent a the conjonction of $W(X_{0a}, i)$.
- If X_0 is purement of dimension n , Y_0 a section hyperplane smooth of X_0 , and que $i < n$, then $W(Y_0, i) = W(X_0, i)$: this resulte of the theorem of Lefschetz faible.

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For prouver the assertions $W(X_0, i)$, one se ramene successivement :

- par c), a supposer X_0 purement d'une dimension n ;
- par b), a supposer of more $0 \leq i \leq n$;
- par a) and d), a supposer of more $i = n$;
- par a) and c), a supposer of more X_0 absolument irreductible.

Ce case is justiciable of (??).

(7.3) Prouvons (??). For all entier k , a_k is value proper of F^* agissant on $H^{kd}(X_k, Q)$ (formule of Kunneth). For k pair, X_k is justiciable of (??), d'ou

$$q^{kd/2-1} \leq |a_k| \leq q^{kd/2+1}$$

and

$$q^{kd/2-k} \leq |a_k| \leq q^{kd/2+k}.$$

Faisant tendre k vers l'infini, one trouve (??).

8. First applications

Theoreme 8.1 (8.1). Let $X_0 \subset P^{n+r}$ F_q a intersection complete non singuliere on F_q , of dimension n and of multidegre $(d_1, \dots, d_r)$. Let b_0 the nieme nombre of Betti of intersections completes nonsingular complexes, of memes dimension and multidegre. Posons $b = b_0$ for n impair, and $b = b_0 - 1$ for n pair. Then

$$|\#X_0(F_q) - \#P^n(F_q)| \leq b \cdot q^{n/2}.$$

Let X/F_q deduit of X_0 , and $Q^i \otimes x_i$ the droite in $H^{2i}(X, Q)$ engendree par the ieme cup-puissance of the cohomology class d'une section hyperplane. Sur cette droite, F^* agit par multiplication par q^i . La cohomologie of X is sum of $Q^i \otimes x_i$ ($0 \leq i \leq n$) and of the partie primitive of $H^n(X, Q)$, of dimension b . After (1.5), il existe therefore b nombres algebriques a_j , the eigenvalues of F^* agissant on cette cohomologie primitive, tels que

$$\#X_0(F_q) = \sum_{i=0}^n q^i + (-1)^n \sum_{j=1}^b a_j.$$

After (??), $|a_j| = q^{n/2}$ and

$$|\#X_0(F_q) - \#P^n(F_q)| = |\sum_{i=0}^n q^i - \sum_{j=1}^b a_j| \leq \sum_{j=1}^b |a_j| \leq b \cdot q^{n/2}.$$

i=0 j=1 j=1

Theoreme 8.2 (8.2). Let N a entier ≥ 1 , $e : (Z/N)^* \rightarrow C^*$ a caractere, k a entier ≥ 2 and f a form modulaire holomorphe on $G_0(N)$, of weights k and of caractere e : f is a

□ fonction
□ holomorphe
a b
c d

on the demi-plan of Poincare, telle que for $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, Z)$ with

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$c \not\equiv 0 \pmod{N}$, one ait

$$f\left(\frac{az+b}{cz+d}\right) = e(d)(cz+d)^k f(z).$$

On suppose f cuspidale and primitive ("new" au sens d'Atkin-Lehner and of Miyake), en particulier

P vecteur propre of operateurs of Hecke T_p ($p \nmid N$). Posons $f = \sum_{n=1}^{\infty} a_n q^n$

, with $f = e^{2\pi i z}$ (and $a_1 = 1$). Then, for p premier ne divisant pas N ,

$$|a_p| \leq 2 * p^{(k-1)/2}.$$

En d'autres termes, the roots of l'equation

$$T^2 - a_p T + e(p) * p^{k-1}$$

are of value absolue $p^{(k-1)/2}$.

Ces roots are en effet of eigenvalues of Frobenius agissant on the H^{k-1} d'une variety projective nonsingular of dimension $k-1$ definie on F_p .

Sous of hypotheses restrictives, ce fait is demontre in mon expose Bourbaki (Formes modulaires and representations p -adiques, expose 355, fevrier 1969, in : Lecture Notes in Mathematics, 179). Le case general n'est pas beaucoup more difficile.

Remarque 8.3 (8.3). J. P. Serre and moi-meme avons recemment of- shows que (??) reste vrai for $k=1$. La proof is all differente.

L'application suivante m'a been suggeree par E. Bombieri.

Theoreme 8.4 (8.4). Let Q a polynome a n variables and of degre d on F_q , Q_d the partie homogene of degre d of Q and $\psi : F_q \rightarrow C^*$ a caractere additif non trivial on F_q . On suppose que :

- (i) d is premier a p ;
- (ii) l'hypersurface H_0 in P^{n-1} definie par Q_d is smooth.

Then

$$\sum_{x_1, \dots, x_n \in F_q} \psi(Q(x_1, \dots, x_n)) \leq (d-1)n * q^{n/2}.$$

Quitte a remplacer Q par a multiple scalaire, one peut supposer (and one supposera) que

$$(8.1) \quad \psi(x) = \exp(2\pi i \text{Tr}_{F_q/F_p}(x)/p).$$

Let X_0 the revetement etale of l'espace affine A_0 of dimension n on F_q d'equation $T^p - T = Q$, and let α the projection of X_0 on A_0 :

$$\alpha : X_0 = \text{Spec}(F_q[x_1, \dots, x_n, T]/(T^p - T - Q)) \rightarrow A_0.$$

Le revetement X_0 is galoisien, of group of Galois Z/p ; $i \in Z/p = F_p$ agit par $T \mapsto T + i$.

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Let $x \in A_0(F_q)$ and calculons l'Frobenius endomorphism of the fibre of X_0/A_0 en x . Posons $q = p^f$ and let F_q a cloture algebrique of F_q . For $(x, T) \in X_0(F_q)$ au-dessus of x , one a $F((x, T)) = (x, T^q)$, and

$$T = T + \sum_{i=0}^{f-1} (T^{p^{i+1}} - T^{p^i}) = T + \sum_{i=0}^{f-1} (Q(x))^{p^i} = T + \text{Tr}_{F_q/F_p}(Q(x)).$$

C'est l'action of l'element $\text{Tr}_{F_q/F_p}(Q(x))$ of the group of Galois.

Let E the corps of roots p -iemes of l'unite and l a place finie of E premiere a p . Nous travaillerons en cohomologie l -adique. For $j \in Z/p$, let $L_{j,0}$ the El -systeme local of rang a on A_0 defini par X_0 and $\psi(-jx) : Z/p \rightarrow E^* \subset Cl^* : \alpha \mapsto L_{j,0}$ and $t(i * x) = \psi(-ji)L_j(x)$. Notons without 0 the objects deduits of A_0 , X_0 , $L_{j,0}$ par extension of scalaires a F_q . La formule of traces (??) for

Lj,0 s'ecrit

$$(8.2) \quad \sum_{x \in F_n} \sum_{q=0}^n \sum_{i=0}^n \text{ps}(Q(x_1, \dots, x_n)) = (-1)^i \text{Tr}(F^*, \text{Hci}(A, L_j)).$$

$$(8.3) \quad \text{On a } a^* \text{El} = \sum_j L_j, 0, \text{ and therefore } \text{Hci}(X, Q^* \otimes Q^* \text{El}) = \sum_j \text{Hci}(A, L_j).$$

For $j = 0$, L_j is the sheaf constant El ; ce facteur corresponds a l'inclusion, par image reciproque, of the cohomology of A in celle of X .

Lemme 8.5 (8.5). (i) For $j \neq 0$, $\text{Hci}(A, L_j)$ is zero for $i \neq n$; for $i = n$, cet space of cohomologie is of dimension $(d-1)n$. (ii) For $j \neq 0$, the cup-produit

$$\text{Hcn}(A, L_j) \otimes \text{Hcn}(A, L_{-j}) \rightarrow \text{Hc}_{2n}(A, \text{El}(-n))$$

is a dualite parfaite.

(iii) X_0 is a ouvert d'une variety projective nonsingular Z_0 .

Deduisons (??) of (??). Let $j_0 : X_0 \rightarrow Z_0$ and $j : X \rightarrow Z$ deduit par extension of scalaires a F_q . After (??), (i), and (??) for Z_0 , il suffices of prouver l'injectivite of

$$\text{Hcn}(A, L_j) \rightarrow \text{Hcn}(X, L_j) = \text{Hcn}(X, \text{El}) \rightarrow \text{H}^n(Z, \text{El}).$$

On a $\text{Tr}(a^* u \cup a^* v) = i \text{Tr}(j_0^! a^* u \cup j_0^! a^* v)$, therefore cette injectivite resulte of (ii).

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(8.6) Prouvons (??) (iii). Let P_0 l'espace projectif on F_q complete of A_0 par adjonction d'un hyperplan a l'infini P_0^∞ , $H_0 \subset P_0^\infty$ d'equation $Q_d = 0$ and Y_0 the revetement of P_0 normalise of P_0 in X_0 .

$$(8.4) \quad \frac{A_0}{P_0} \rightarrow \frac{Y_0}{P_0} \rightarrow \frac{H_0}{P_0^\infty}$$

Etudions Y_0/P_0 pres of l'infini, localement for the topologie etale.

Lemme 8.6 (8.7). Y_0 is smooth en dehors of l'image reciproque of H_0 .

Le diviseur of the fonction rationnelle Q on P_0 is sum of sa partie finie $\sum (Q)_f$ and of $(-d)$ fois l'hyperplan a l'infini. On a :

$$\sum (Q)_f = \sum (Q)_f - d * P_0^\infty, \quad \sum (Q)_f \cap P_0^\infty = H_0.$$

A distance finie, $Y_0 = X_0$ is etale on A_0 , therefore smooth. A l'infini, mais en dehors of l'image reciproque of H_0 , il existe of coordonnees locales $(z_1, \dots, z_n)$ telles que $Q = z_1 - d$ (this utilise que $(d, p) = 1$). Dans ces coordonnees, Y_0 apparait as product d'une courbe and d'un space smooth (correspondant aux coordonnees $z_2, \dots, z_n$). Par normalite, il is smooth.

Lemme 8.7 (8.8). Au voisinage etale d'un point au-dessus of H_0 , Y_0 is smooth on a surface singuliere normale, always the meme.

On peut cette fois trouver of coordonnees locales telles que $Q = z_1 - d * z_2$. En effet, puisque H_0 is smooth, $\sum (Q)_f$ is smooth au voisinage of l'infini and coupe P_0^∞ transversalement. Cette form is independante of the point choisi, and n'utilise que deux coordonnees, d'ou l'assertion. (8.9) On sait que the procede suivant (dit a Zariski) permet of resoudre the singularites of surfaces : alternativement, one normalise and one eclate the lieu singulier (reduit). Les operateurs en jeu commutent a the localisation etale and au product par a space smooth. Le procede of Zariski permet therefore again of resoudre the singularites d'un space qui, as Y_0 , is, localement for the topologie etale, smooth on a surface. La resolution obtenue of Y_0 is the Z_0 cherche.

If T is a courbe on a surface S , contenant the lieu singulier, and que T_0 is l'image reciproque of T in the resolution of Zariski S_0 of S , one sait que if one eclate of facon iteree in S_0 the lieu singulier (reduit) of $(T_0)_{\text{red}}$, one obtains a surface S_{00} telle que l'image reciproque reduite $(T_{00})_{\text{red}}$ of T in S_{00} let a diviseur a croisements normaux.

A nouveau, the operations en jeu commutent a the localisation etale and au product par a space smooth. Raisonnant as more haut, and observant que $(Y_0, \text{l'infini})$ is localement smooth on a (S, T) , one peut meme trouver Z_0 tel que $Z_0 - X_0$ let a diviseur a croisements normaux.

(8.10) Prouvons (??) (i), (ii). Ce are of assertions geometrics ; this we permet of travailler desormais on F_q . Let $S \subset \mathbb{A}^n$ l'espace affine on F_q qui parametriser les polynomes a n variables of degree $\leq d$, and let S l'ouvert of $S \subset \mathbb{A}^n$ correspondant aux polynomes dont the partie homogene of degree d is of discriminant non zero. On note $Q_S \in H^0(S, \mathcal{O}_S[x_1, \dots, x_n])$ the polynome universel on S , and X_S the revetement galoisien etale of $A_S = A_n \times_S d'equation T^p - T = Q_S$, of group of Galois Z/p . Let $P_S = P_n \times_S$ l'espace projectif on S complete of A_S and Y_S the normalise of P_S in X_S . On dispose, on S , d'un diagramme analogue a (??).

Les expressions of Q en coordonnees locales given en (??) and (??) restent valables in the situation presente, with parametres, of sorte que, localement for the topologie etale on Y_S , Y_S/S is isomorphe au product of S (qui is smooth) with a fibre. La methode canonical of resolution utilisee en (8.9) we fournit a compactification relative Z_S/S of X_S/S , with $Z_S - X_S$ a diviseur a croisements normaux relatifs on S

$$X_S \quad / \quad Z_S$$

$$\begin{array}{ccc} \square & & \square \\ AS & & /S \\ & \searrow & \\ & \{\{\} & \\ & \{\{\} & \\ & \{\{\} & \\ & \}\} & \\ \square \{ & & \\ S & & \end{array}$$

(f proper and smooth, u plongement ouvert, $Z_S - X_S$ diviseur a croisements normaux relatifs).

Let $L_{j,S}$ le $\mathcal{O}_S$ -sheaf on A_S deduit as en (??) of X_S/A_S . On a $a^* \mathcal{E}_l = j_* L_{j,S}$, therefore

$$R_i(f_u)! \mathcal{E}_l = \sum_j^M R_i f! (u^* L_{j,S}).$$

Les properties of Z_S assurent que the $R_i(f_u)! \mathcal{E}_l = R_i f! (u^* \mathcal{E}_l)$ are of sheaves localement constants on S . Des lors, the $R_i f! (u^* L_{j,S})$ egalement are localement constants.

Puisque S is connexe, il suffices of prouver (??) (i), (ii) for a polynome Q particulier. On prendra $Q = \sum x_i^d$. Ce polynome satisfies the condition of non-singularite car $(d, p) = 1$. For ce polynome, the variables se separant in the sum exponentielle of (??). Ceci correspond au fait que L_j is the product tensoriel of images reciproques of sheaves analogues $L_{j,i}$ on the facteurs of dimension 1 of $A = A_n$. Par the formule of Kunneth

$$H^0_c(A, L_j) = \sum_i H^0_c(A_1, L_{j,i}).$$

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Ceci ramene the proof of (??) (i), (ii) au case or $n = 1$ and or Q is x^d .

(8.11) Traitons ce case particulier. Le revetement X of A is irreductible, of sorte que for $i = 0, 2$

$$\begin{array}{c} \sim \\ H^i_c(A, \mathcal{E}_l) \rightarrow H^i_c(X, \mathcal{E}_l). \end{array}$$

For $i \neq 0$ and $j \neq 0$, one a therefore $H^i_c(A, L_j) = 0$.

L'assertion (ii) resulte of (2.8) or (2.12) and of ce que $L_j \otimes L_{-j} = \mathcal{O} = \mathcal{E}_l$. For prouver (i), il reste a verifier que

$$kh_c(A, L_j) = 1 - d.$$

After the formule d'Euler-Poincare (voir l'expose Bourbaki 286 of fevrier 1965, par M. Raynaud), this is equivalent au lemma suivant. Lemme 8.8 (8.12). Le conducteur of Swan of L_j a l'infini is egal a d .

Cet enonce is equivalent au suivant.

Lemme 8.9 (8.13). Let k a finite field of caracteristique p , $y \in k[[x]]$ a element of valuation d premiere a p , L l'extension of $K = k((x))$ engendree par the roots of $T^p - T = y^{-1}$ and kh the caractere suivant of $\text{Gal}(L/K)$ a values in Z/p :

$$kh(s) = sT - T.$$

Then, kh is of conducteur $d + 1$.

Par extension of the residue field, one se ramene a supposer k algebriquement clos plutot que fini, and one applique : J. P. Serre, Sur the corps locaux a residue field algebraically closed, Bull. Soc. Math. France,

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